Ayan Ahmed

Software Engineer, Google

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Education

2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78

2015 **XII - Karnataka Board,** *KLE Society's Independent PU College, Bangalore* 97.30 %

2013 **X - ICSE,** *B P Indian Public School, Bangalore* 96.33%

Professional Experience

Dec 2022 - Present **Software Engineer, Level IV, Google LLC, New York**
- Working on Keep, a notetaking editor in Google Workspace.

Aug 2020 - Nov 2022 **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
- Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
- Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
- Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.

May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.

May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis,* ***Language Technologies Institute, Carnegie Mellon University***
Machinated a novel technique for paraphrase generation using the von Mises-Fisher (vMF) Loss on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by *Prof. Yulia Tsvetkov*.

May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
Research Intern, ***Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore***
Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by Prof. Veni Madhavan.

Publications and Patents

[Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan*, Sachin Kumar, John Wieting, Yulia Tsvetkov*
In **Workshop on Multilingual Representation Learning**,
The 2021 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2021**)

[Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In **ACM CODS-COMAD 2020** - Young Researchers' Symposium

[Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
In ACM User Interface Software and Technology Symposium 2019 (**ACM UIST 2019**)

[Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, ***Monisha Jegadeesan****, Amrit Singhal, Kush Kumar Singh, Kundan Krishna*
Filed at the US PTO (Application Number: 16/247,235)

[Publication and Poster] **Leveraging Ontological Knowledge for Neural Language Models** (**Paper**, **Poster**) *Ameet Deshpande*, ***Monisha Jegadeesan***
In **ACM CODS-COMAD 2019** - Young Researchers' Symposium

Projects

Paraphrase Generation System (2020)

Built a transformer-based paraphrase generation system using vMF loss, achieving superior diversity scores compared to standard log-likelihood models.
Tech: Python, PyTorch, HuggingFace Transformers

AR Text-to-Scene Composer (2019)

Developed a mobile AR app that converts natural language descriptions into 3D scenes using object size/position prediction from textual features.
Tech: Unity, ARKit, Python, NLP

Cognitive Text Parser (2017)

Created a cognitive text parser combining syntactic and semantic approaches to produce structural representations for downstream NLP tasks.
Tech: Python, NLP, Stanford CoreNLP

Courses :

Natural Language Processing • Machine Learning • Deep Learning
Data Structures and Algorithms • Computer Vision • Distributed Systems
Advanced Algorithms • Compiler Design

Skills :

Languages: Python, Java, C++, JavaScript, TypeScript
Frameworks: PyTorch, TensorFlow, React, Node.js
Tools: Git, Docker, Kubernetes, GCP, Web Assembly, Emscripten
NLP / ML: Transformers, BERT, GPT, HuggingFace, spaCy, scikit-learn
Other: j2Cl, Bazel, Protocol Buffers, REST APIs